GRADUATION PROJECT · FINAL PRESENTATION

Shelfmark

An AI-powered smart e-commerce management system: an Arabic-first commerce platform with seven AI modules, back-office automation, and interactive BI — implemented, measured, and honestly bounded.

Next.js · FastAPI · PostgreSQL · MongoDB Scikit-learn · PyTorch · OpenAI Docker

Every screen in this deck was captured from the running system. All data is synthetic and labelled as such in the product.

1 · PROBLEM AND OBJECTIVES

Three objectives, seven smart modules, one back office

- **OBJ-01** — personalised recommendations and shopping assistance in the customer journey.
- **OBJ-02** — automate inventory, sales operations, and pricing from the same order records.
- **OBJ-03** — interactive BI over consumer behaviour and financial performance.

> Scope note: this is the specified commerce + AI scope — not a general ERP. The original seven modules are preserved, not narrowed.

MODULE	CAPABILITY
SF-01	Hybrid collaborative + content recommendations
SF-02	Grounded assistant for inquiries, choice, shipments
SF-03	Image search, voice search, dynamic facets
BO-01	Demand forecast and automatic reordering
BO-02	Dynamic pricing from supply, demand, rivals, season
BO-03	RFM customer segmentation and campaign export
BO-04	Trained fraud detection before completion
BI-01	Consumer and financial dashboards with reconciliation

2 · ARCHITECTURE

Two surfaces, two sessions, explicit ownership



> PostgreSQL owns price, stock, and orders. MongoDB holds descriptive content only. One canonical UUID links both stores, so a product can never have two prices.

- **34 API routes** across catalog, commerce, identity, risk, and back office.
- **28 PostgreSQL tables** across eight migrations; MongoDB validator and indexes.
- **Separate customer and operator sessions**: argon2id demo credentials, revocable database sessions, and two independent `httpOnly` cookies; catalogue browsing stays public.
- **Capability scopes** (`risk:review`, `pricing:write`, ...) instead of a role tree.
- **Typed failures**: 404 for unknown, 503 for an active record whose other side is missing.
- **Docker** packaging for API, stores, jobs, and artifacts.

3 · WHAT WAS BUILT

Every module carries a measured number

MODULE	EVIDENCE TO SHOW	STATUS
SF-01 recommendations	nDCG@10 **1.0** vs popularity 0.690 (+44.83%)	verified on synthetic
SF-02 assistant	Grounded FAQ + catalog + own shipments; cross-customer denial	provider pending
SF-03 search	Image top-1 **1.0**, recall@3 **1.0**; facets and combined filters	image verified
BO-01 inventory	Holdout MAPE **3.35%**; 13 forecasts, 12 reorders, 0 duplicates on rerun	verified on synthetic
BO-02 pricing	±20% cap, cost+5% floor, 24h freshness, immutable audit	verified on synthetic
BO-03 segmentation	**38** classified, **2** without history, six segments, CSV export	verified on synthetic
BO-04 fraud	Recall **83.3%** at **7.8%** FPR, before order completion	verified on sandbox
BI-01 dashboards	Consumer + financial KPIs, filters recompute, sources reconcile	verified on synthetic

"Verified on synthetic" means the pipeline, the policy, and the integration are proven — not that commercial quality is claimed. Section 6 states each limit.

4 · SF-01 RECOMMENDATIONS

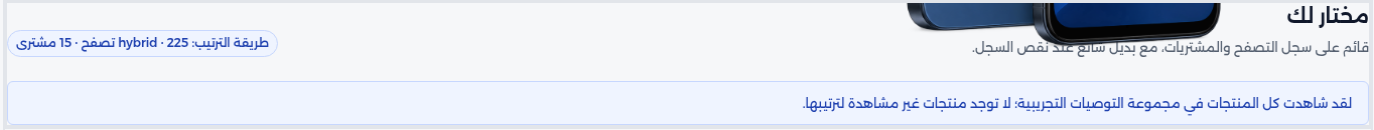
Ranked from real browsing and purchase history — and honest when exhausted

1.0	0.690	+44.8%
hybrid nDCG@10 (holdout)	popularity baseline	relative lift

- Collaborative similarity from item co-occurrence, content similarity from product text, combined 60/40 with a popularity tie-breaker.
- Signals come from the application's own records: browsing events and confirmed purchases.
- Seen items are excluded — the rail on the right shows what happens when the customer has seen everything.

> The seeded demonstration customer has already seen all four products covered by the trained artifact, so the rail states that instead of inventing a fallback. The API run and the recorded evaluation carry the ranking evidence.

Product page · recommendation section with the exhausted state




```
GET /v1/recommendations?limit=5
{
  "strategy": "hybrid",
  "basis": { "browsing_events": 226, "purchased_items": 15 },
  "items": []
}
```

Seen items are absent from `items`; with no unseen candidate the response is empty and the rail says so.

5 · SF-02 / SF-03 ASSISTANT AND SEARCH

Grounded answers, image retrieval, and honest failure states

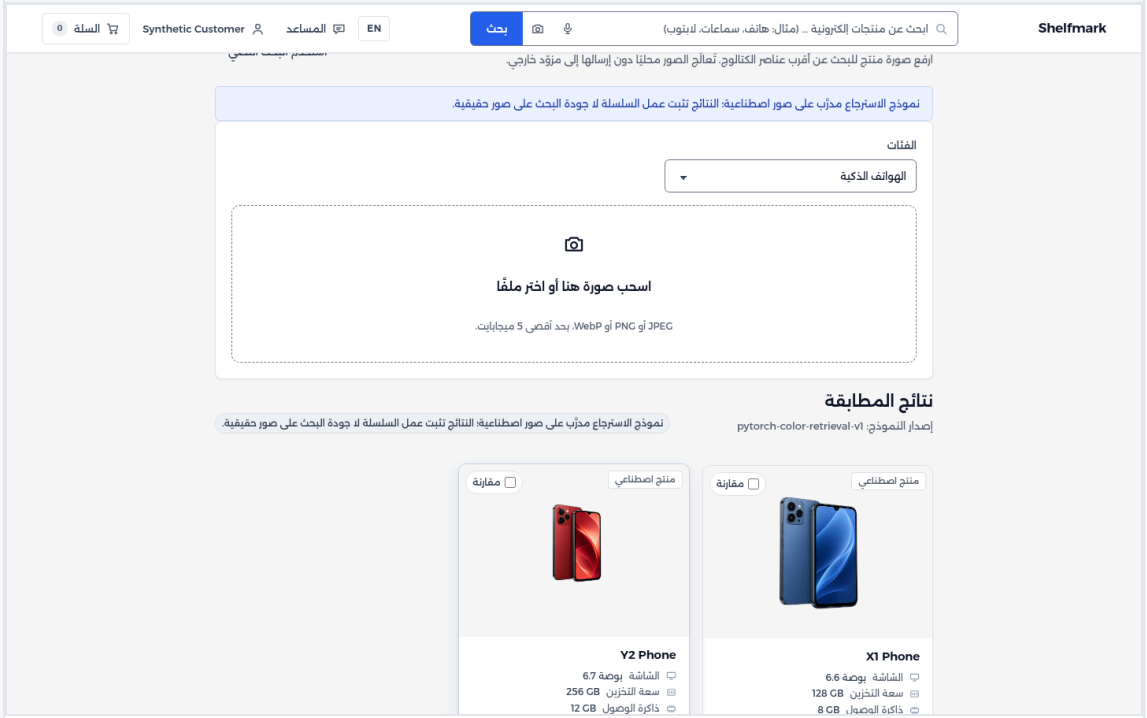
SF-02 · Assistant

- Grounded in project-authored bilingual FAQ, the live catalog, and the customer's own shipments.
- Cross-customer shipment lookup is denied; other customers' records never appear.
- Without a provider key the surface returns a typed **503** instead of inventing an answer.

SF-03 · Search

- PyTorch colour-feature retrieval: top-1 **1.0**, recall@3 **1.0** on the synthetic set.
- Dynamic facets combine with the image query and with catalog filters.
- Voice transcription targets `gpt-transcribe`; the provider call is pending a key.

Image search · Phones filter applied, X1 ranked first



6 · BO-01 PREDICTIVE INVENTORY

Forecast → reorder → supplier acknowledgement, without duplicates

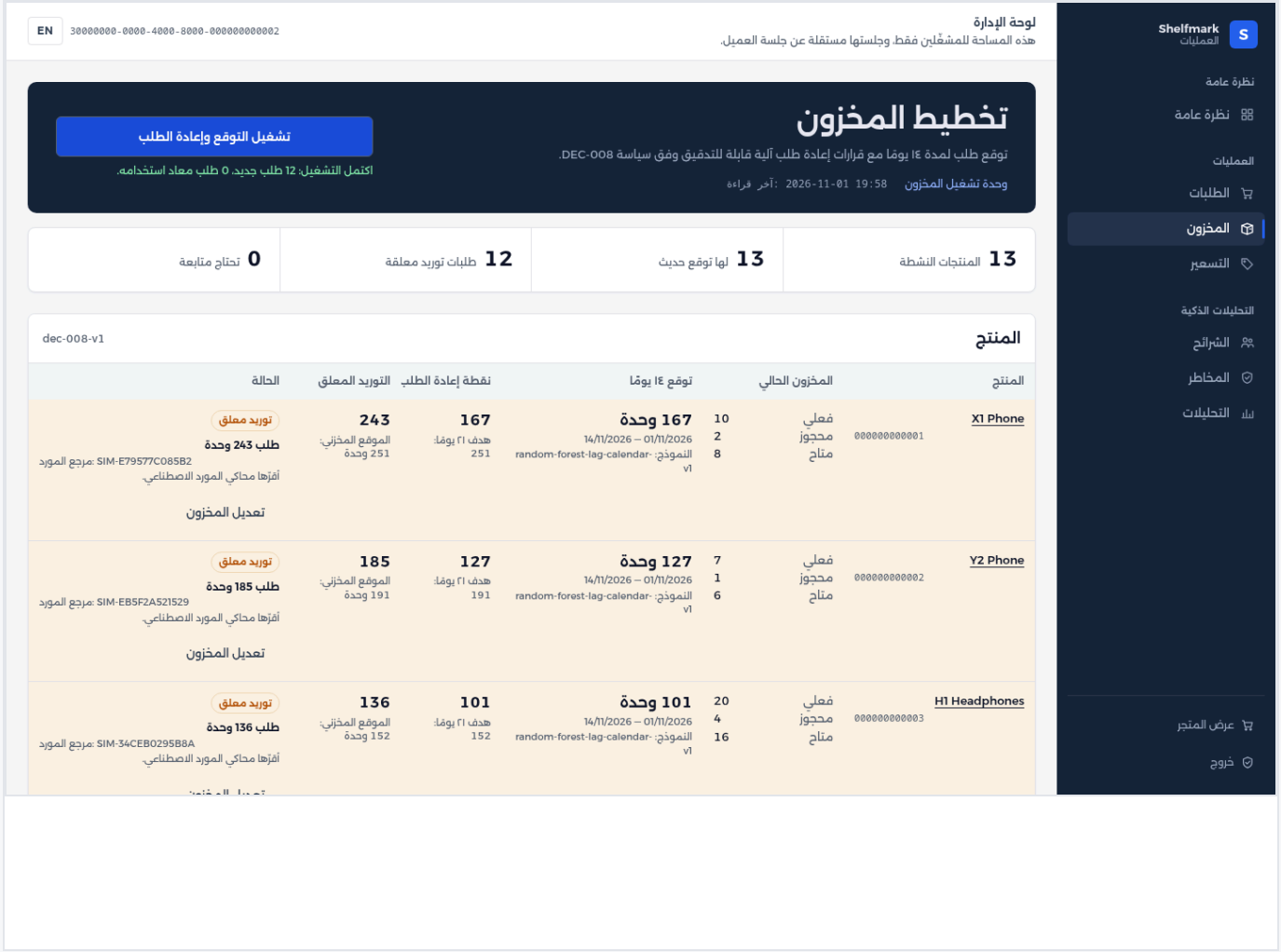
3.35%
MAPE, non-zero demand, final 28-day holdout

12
reorders created on the first run

0
duplicates when the run is repeated

- Leakage-safe lag, rolling, and calendar features; a chronological holdout, never a random split.
- DEC-008 policy: 14-day forecast, 7-day lead time, 7-day safety stock, reorder to 21-day cover.
- Pending replenishment counts toward inventory position, so a live reorder is not re-ordered.
- The supplier is a simulator: failures are preserved for retry and reruns are idempotent.

Back office · inventory run (12 new reorders, 0 reused)



7 · BO-02 DYNAMIC PRICING

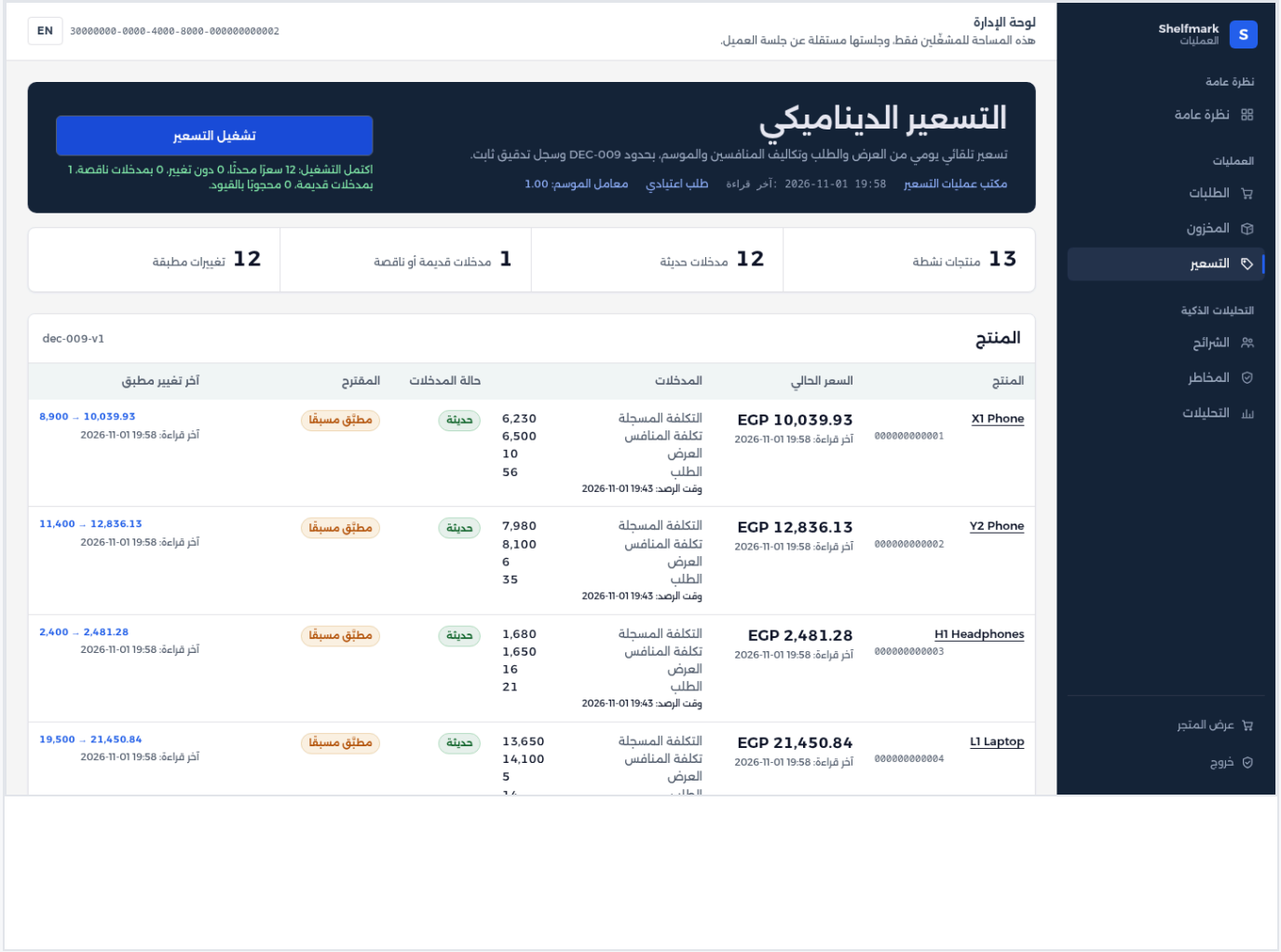
Bounded, evidenced, and never retroactive

- DEC-009: demand relative to supply, competitor cost, and a dated season calendar.
- At most **±20%** per run and never below recorded cost + 5%.
- Inputs older than 24 hours block a change; one observation can never produce two changes.
- Every change stores the full input snapshot as an immutable audit row.

> Rehearsal run: X1 Phone **8,900** → **10,039.93** after the first run, then → **12,047.91** after a new observation with a much higher competitor cost. The second move is +19.98% — inside the 20% cap.

> The order placed before the change still shows **8,900**. Order prices are snapshots; repricing never rewrites history.

Back office · pricing run with inputs, proposals, and audit trail



8 · BO-03 RFM SEGMENTATION

Six documented segments from a reproducible window

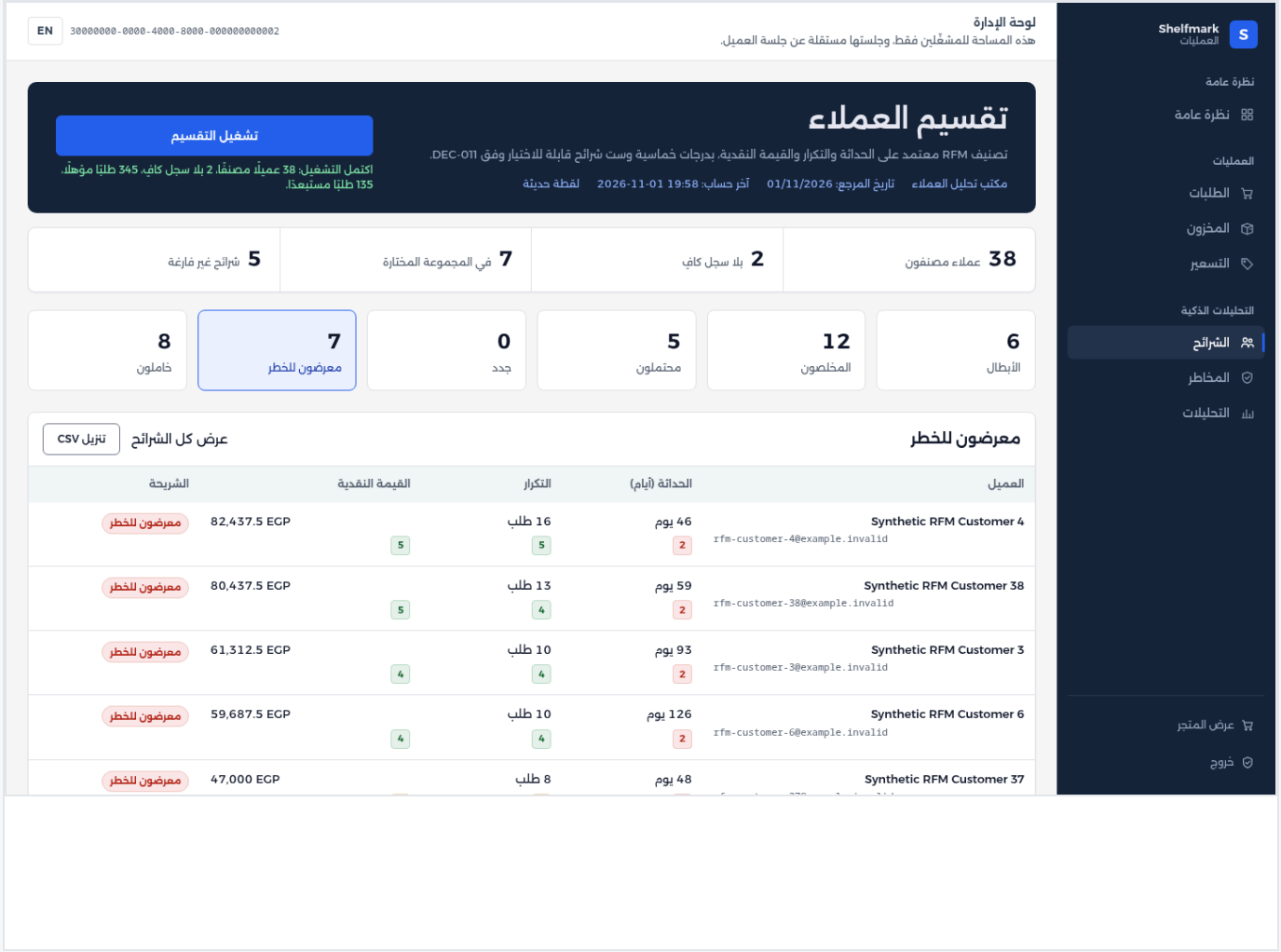
38
customers classified

2
counted as insufficient history, not scored zero

6
selectable segments with a CSV export

- DEC-011: rolling 365-day window, quintile scores on recency, frequency, and monetary value.
- Eligible orders are paid or completed minus refunds; cancelled and failed orders are excluded.
- Clicking a segment filters the members; the CSV row count matches the selected group exactly.
- The policy matrix beside the table shows the rules, so a score is always explainable.

Back office · At Risk selected, 7 members in the group



9 · BO-04 FRAUD DETECTION

Risk is assessed before an order can complete

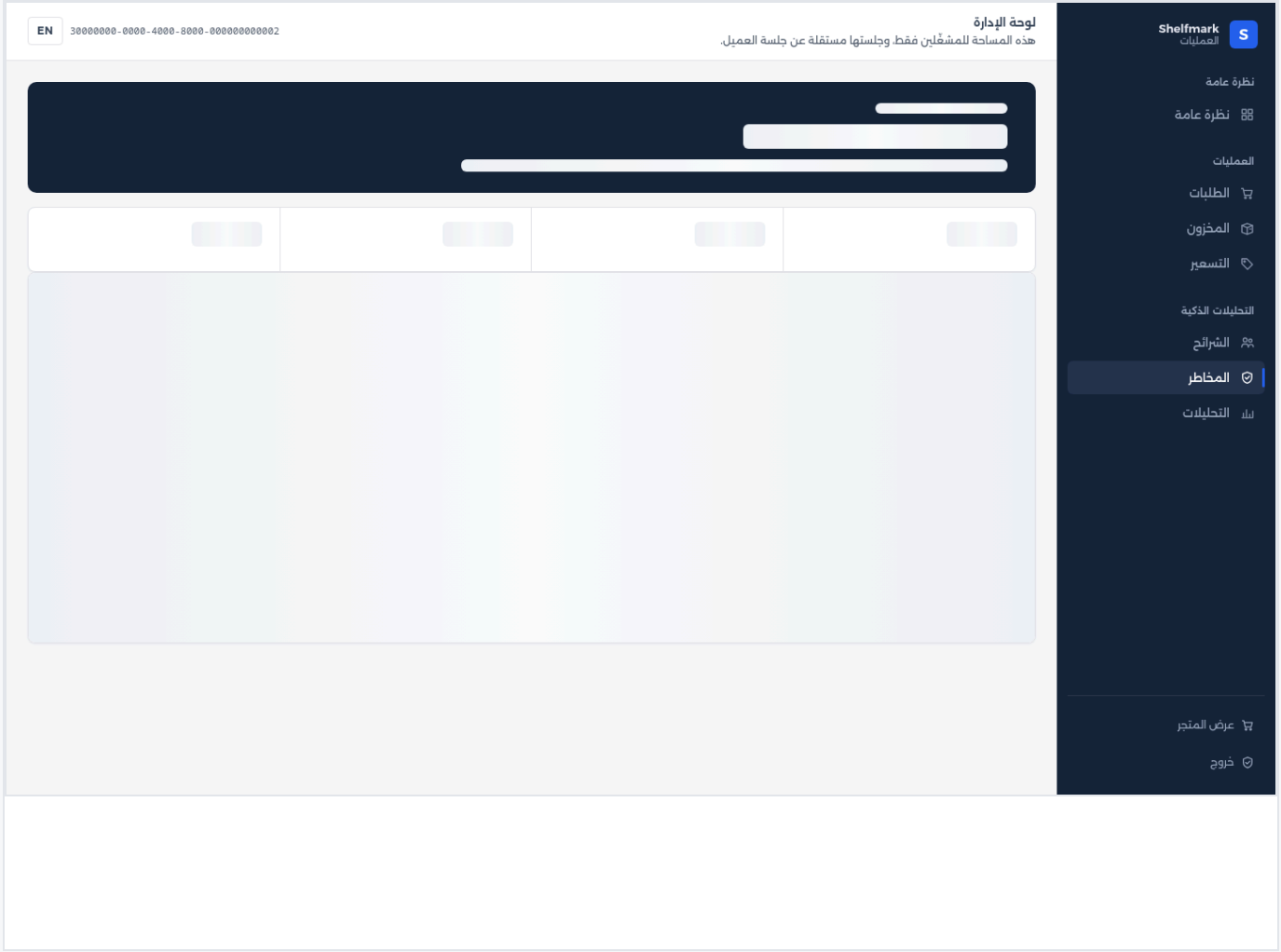
83.3%
recall on a sealed holdout

7.8%
false-positive rate

0.699
review threshold

- Logistic baseline compared against histogram gradient boosting; the threshold is selected on validation and sealed before the holdout.
- A flagged or failed assessment sends the order to review — **no order completes first**.
- Allow and decline are operator decisions, recorded once, with score, model version, and reasons.
- Missing model artefacts fail closed to review, never to approval.

Back office · review queue, model card, and audited decisions



10 · BI-01 DASHBOARDS

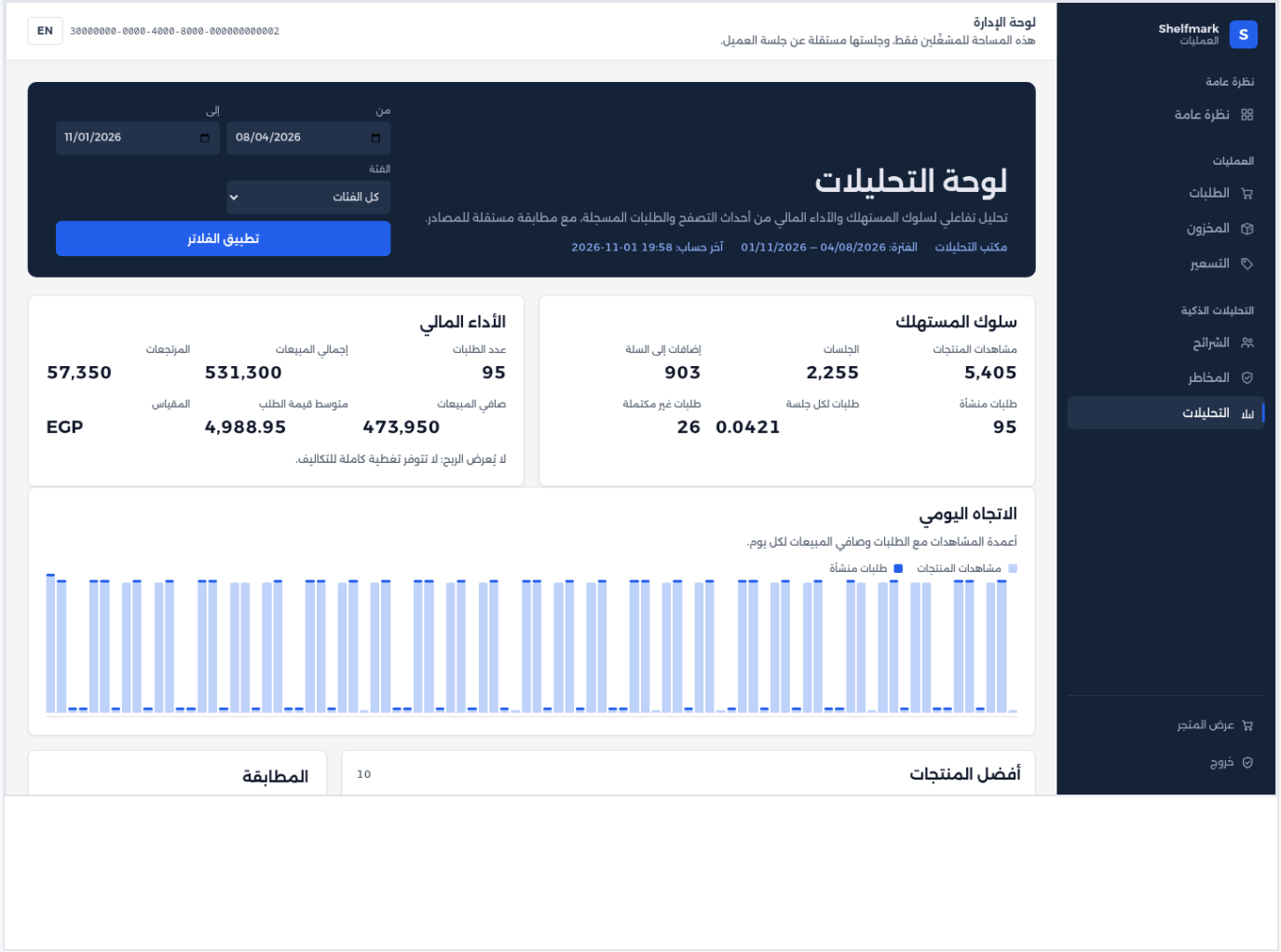
Interactive analysis that reconciles against its own sources

- Consumer panel: views, sessions, add-to-carts, created orders, orders per session, unfulfilled orders.
- Financial panel: order count, gross sales, refunds, net sales, average order value.
- Date and category filters recompute every metric; the phones filter reduces views and revenue and still reconciles.
- Reconciliation compares reported totals with values recomputed from order lines — a mismatch is shown, not hidden.

> **Profit is not displayed.** There is no complete cost coverage, so a profit figure would be an invention.

> Rehearsal window (90 days): 5,405 views · 2,255 sessions · 94 orders · EGP 524,050 gross · EGP 466,800 net.

Back office · analytics, Arabic RTL



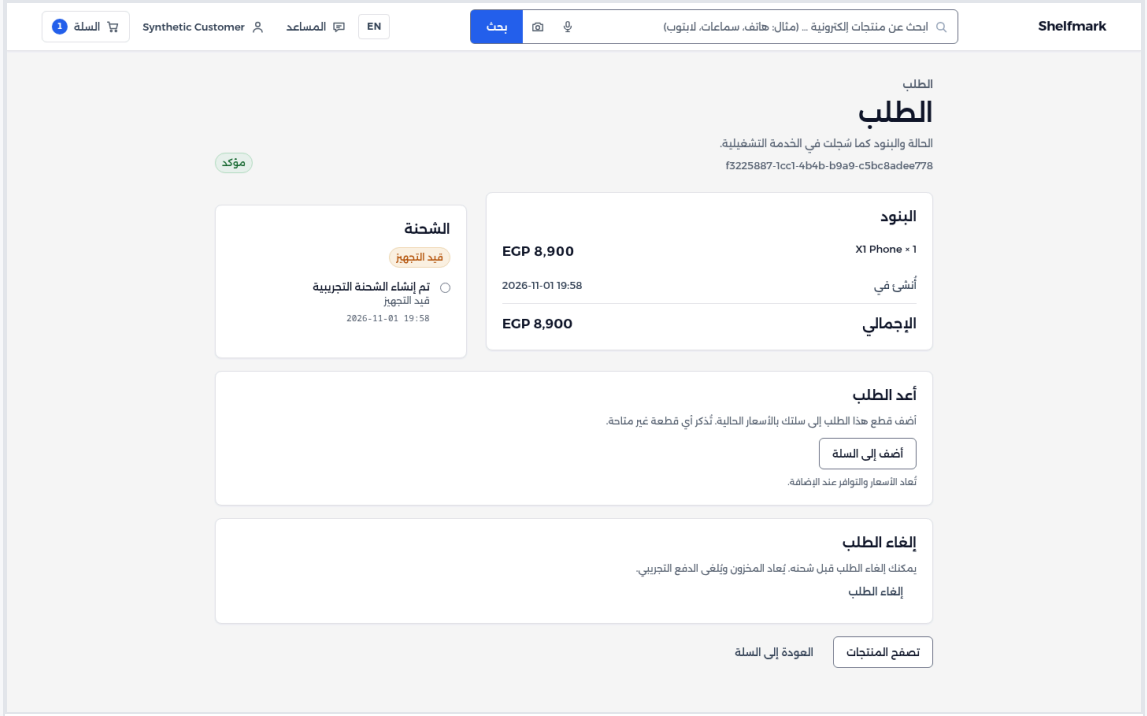
11 · COMMERCE SPINE

Checkout, risk, and history that cannot be rewritten

- **Exact EGP totals** from decimal storage, never floats.
- **Current stock and price** are checked at checkout; a changed cart is refused.
- **Idempotent checkout**: ten simultaneous retries of one key created exactly one order.
- **Immutable order snapshots**: repricing never changes a placed order.
- **Ownership-scoped reads**: another customer's order returns 404 with no data.

> A confirmed order creates a shipment with bilingual events, visible only to its owner. The operator queue finds that order under `orders:read`, and the same page advances its shipment, refunds, or cancels it under `orders:write` — every action attributed to the operator.

Customer order · confirmed, with its shipment timeline



12 · ENGINEERING DECISIONS

Four decisions worth defending

- **DEC-007 — competitors' costs.** The source wording is preserved literally; with no real feed, the values are imported synthetically and labelled, not quietly reinterpreted.
- **DEC-009 / DEC-011 — policies before code.** Pricing bounds and the RFM window are fixed decisions, so the same input always produces the same output.
- **Idempotency everywhere.** Checkout keys, reorder keys, one change per pricing observation, and one RFM snapshot per customer per date.
- **Fail closed, state honestly.** A missing artefact is a typed 503 or a review — never a silent pass. Stale inputs block a price change. Insufficient history is counted, not scored zero. Profit is omitted without cost coverage.

13 · TESTING AND QUALITY

Stage 4 evidence, mapped requirement by requirement

SUITE	RESULT
API unit and route tests	**143 passed**
Contract and database tests	**18 passed** (PostgreSQL 18.6, MongoDB 8.0.34)
Black-box integration flows	**9 passed**
Security and isolation checks	**7 passed**
Workload checks (10 req/s, 100 sessions)	**3 passed**
axe-core WCAG 2.2 AA (46 pages, AR + EN)	**0 violations**

> Rehearsing the demo script caught a production-build defect that unit tests could not see: every back-office server action failed because a `"use server"` module exported a state object. Fixed, rebuilt, re-rehearsed to zero drift.

PATH	TARGET	MEASURED P95
Catalog list	≤500 ms	**8.5 ms** sustained · 156.6 ms at 100 sessions
Recommendations	≤500 ms	**33.8 ms** · 252.6 ms
Analytics (month)	≤2 s	**30.3 ms**
Checkout with risk	≤2 s	**≤330 ms**

Full report: `docs/test-report.md`. Rehearsal log: `docs/presentation/rehearsal.md`.

14 · LIMITATIONS

What this system is not

- **No live AI provider proof.** The assistant and voice surfaces are verified with an injected provider; no `OPENAI_API_KEY` was available, so both return a typed 503 by design.
- **Synthetic data only.** Every model, dashboard, and evaluation uses synthetic or sandbox records. Nothing here is a commercial claim.
- **No cloud deployment.** Docker packaging and the deployment runbook are verified; the AWS placement needs account access and has not been executed.
- **Accessibility is automated only.** axe-core and a keyboard smoke test pass; no manual screen-reader session was run.
- **Authentication is a labelled demo harness.** Local `.invalid` accounts, argon2id hashes, and revocable database sessions; there is no OAuth/OIDC, email verification, email-based password reset, or MFA, and no production identity-provider claim.
- **The storefront is deliberately Arabic-first RTL.** It recorded 58.53% against the approved LTR reference instead of the 72% target; that mismatch is documented rather than restated as a pass.
- **The recommendation artifact covers four products**, so the seeded customer has nothing unseen left to rank — the rail says so instead of faking a result.

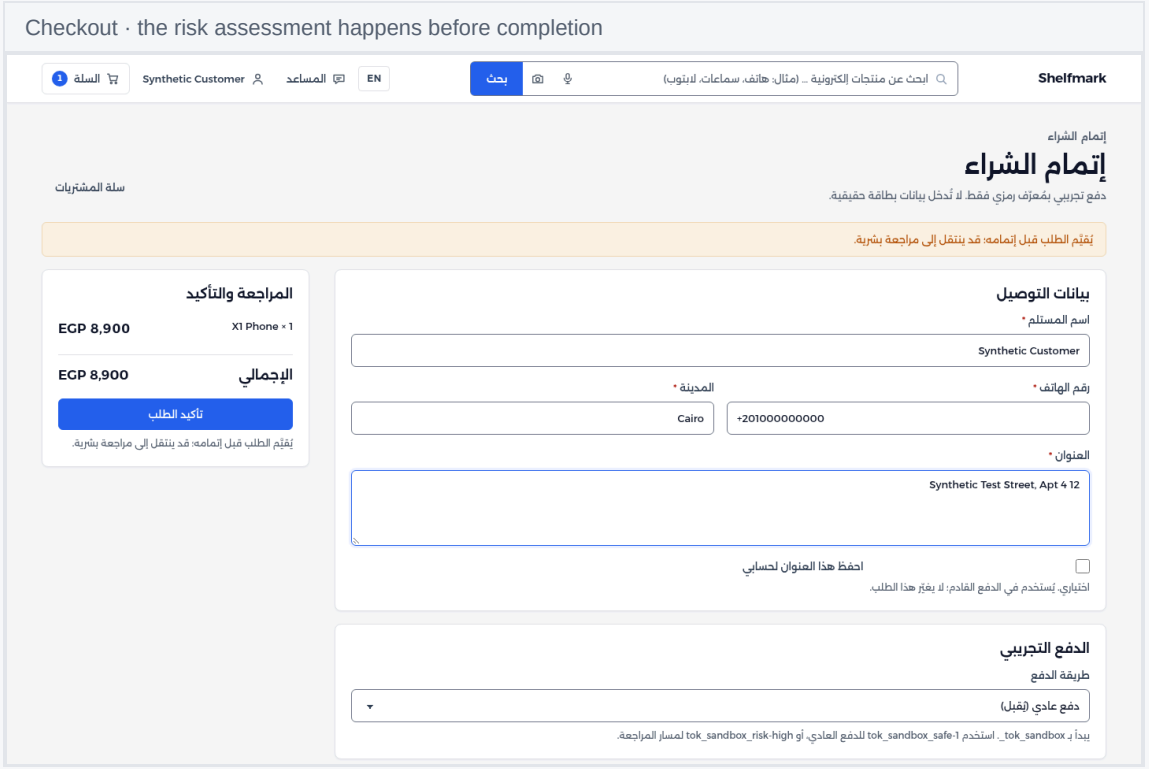
15 · DEMO

One flow, every capability

1. **Sign in** — customer at `/login`, operator at `/backoffice/login`.
2. **Discovery** — filter, image search, and an honest rejection message.
3. **Recommendations** — the exhausted state, then the API evidence.
4. **Checkout** — normal payment confirms; a high-risk token goes to review; the operator allows it, opens it in `/backoffice/orders`, and advances its shipment.
5. **Inventory** — run the forecast twice; the second run creates nothing.
6. **Pricing** — run it, show the audit, then show that a placed order keeps its price.
7. **Segmentation** — run it, filter a segment, export the CSV.
8. **Analytics** — apply a category filter and read the reconciliation.

> The rehearsal script drives this exact sequence against a freshly seeded stack and finished with **zero drift** across **26 captured screens**.

Checkout · the risk assessment happens before completion



CONCLUSION

All seven modules and BI are implemented and evidenced

What remains is access — a provider key, a cloud account, a manual accessibility pass — not missing functionality.

Source: `README.md` Evidence: `docs/test-report.md` Rehearsal: `docs/presentation/rehearsal.md`

Reproducible from the repository: Docker Compose for the stack, `uv` for the suites, `npm` for the storefront and the accessibility gate.